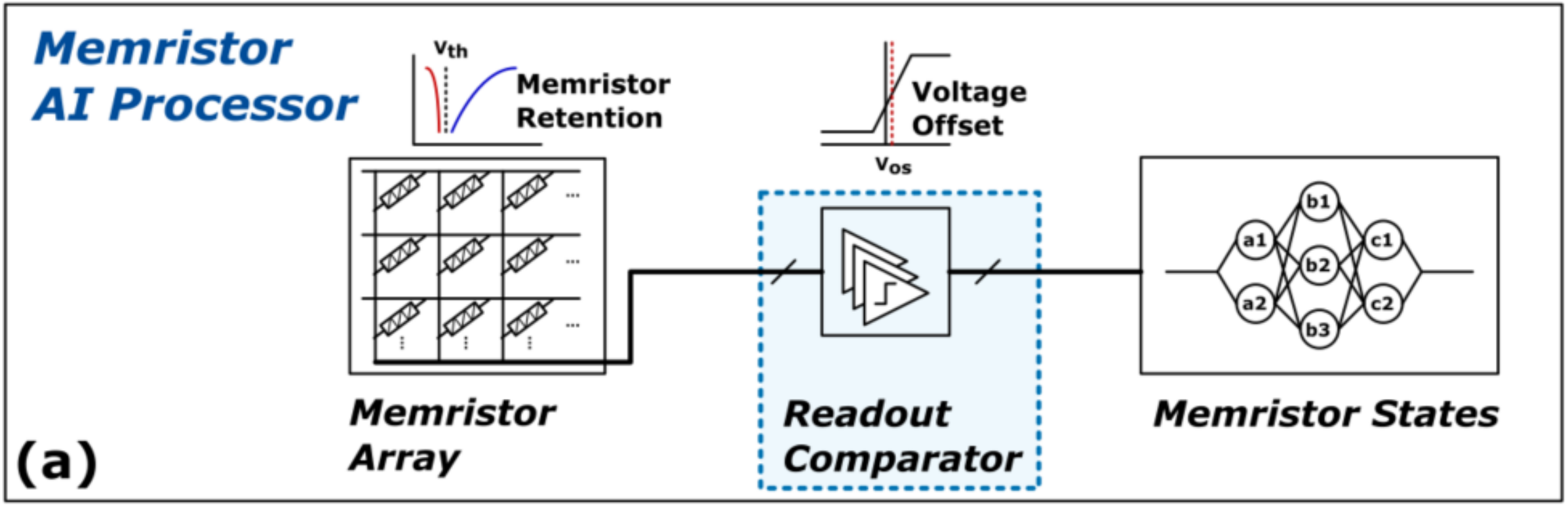
A Clock-Independent, Time-Domain Rapid Calibration Method for Memristor-based Analog Computing AI Processors

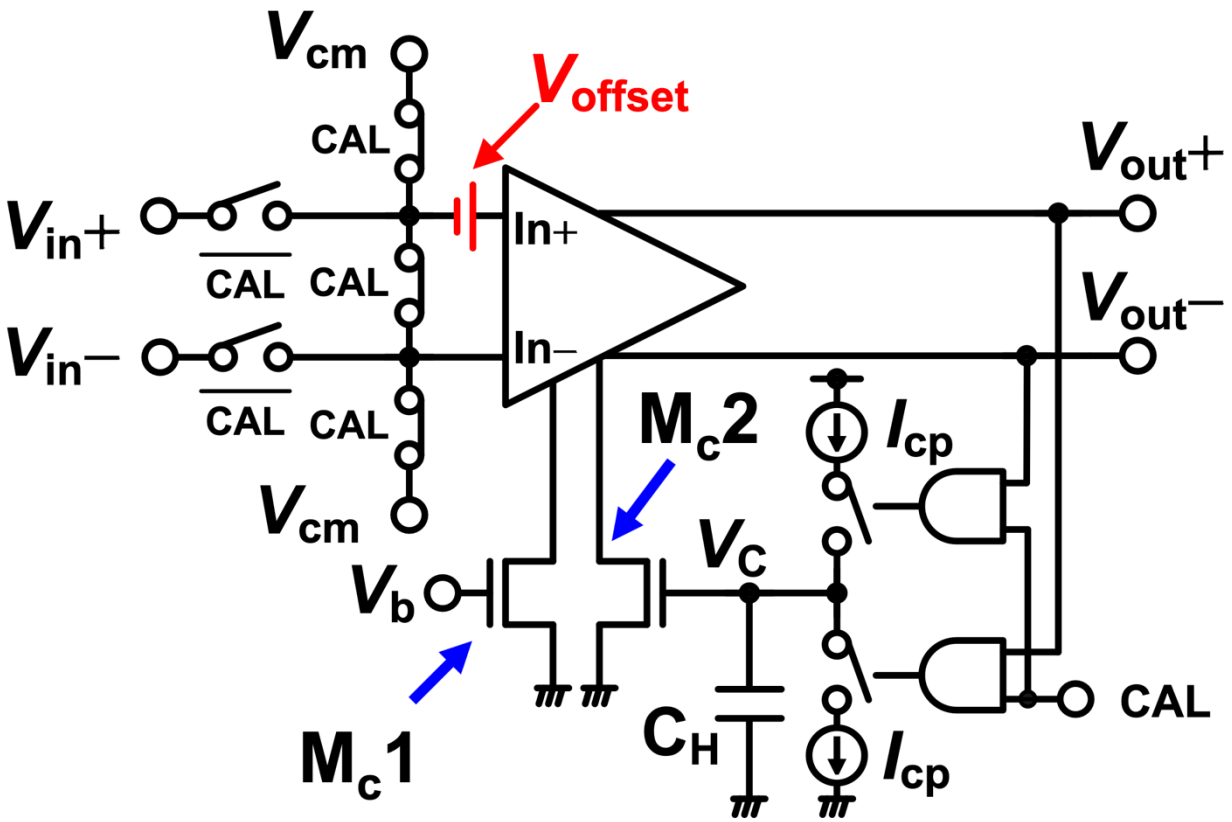
Chen, Junzhe (JC)

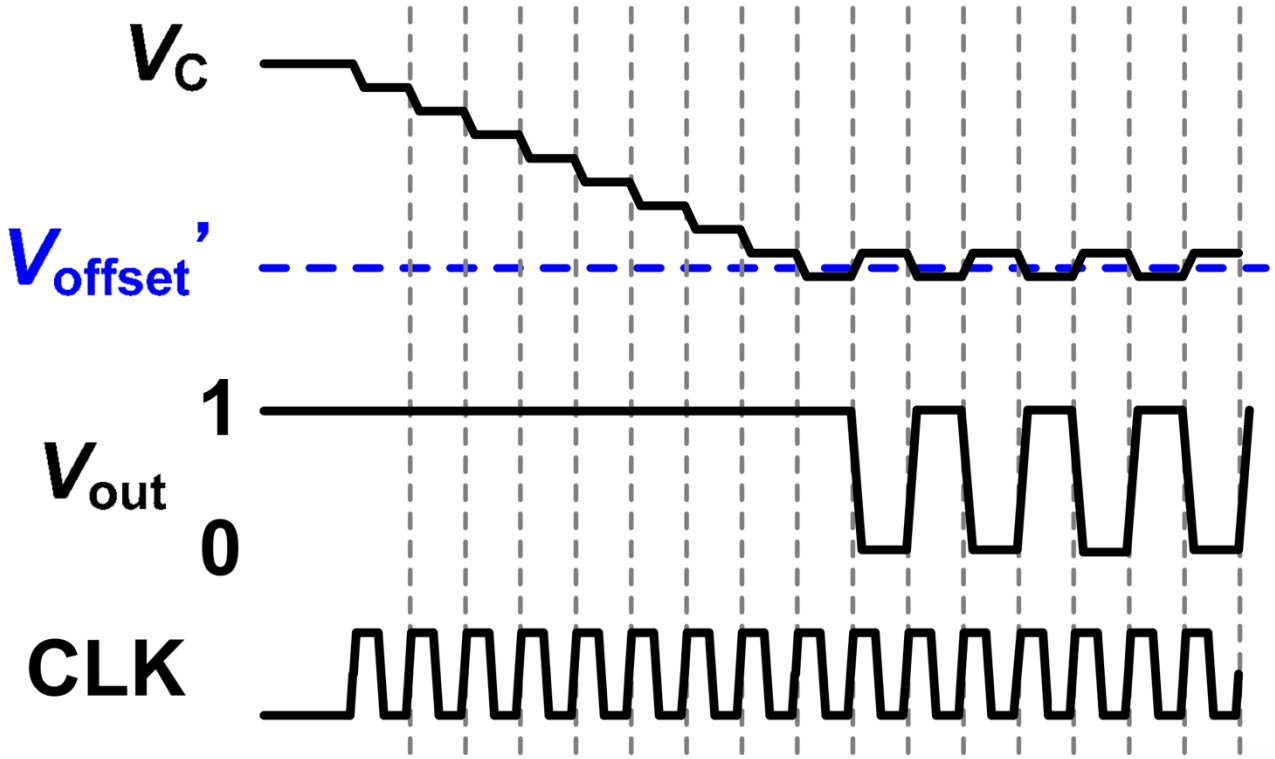
20/Oct/2026, Bristol, UK

System

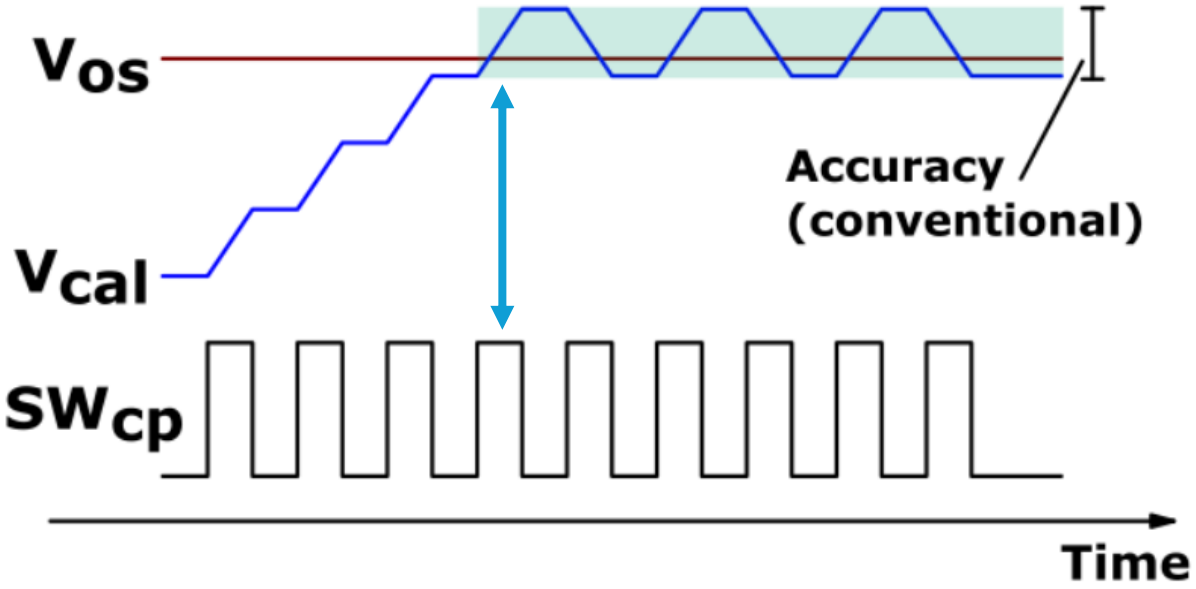

Memristor
AI Processor
Vth
Memristor
Retention
Voltage
Offset
Vos
Memristor
Array
Readout
Comparator
Memristor States
a1
a2
b1
b2
b3
c1
c2
(a)

Miyahara [1]

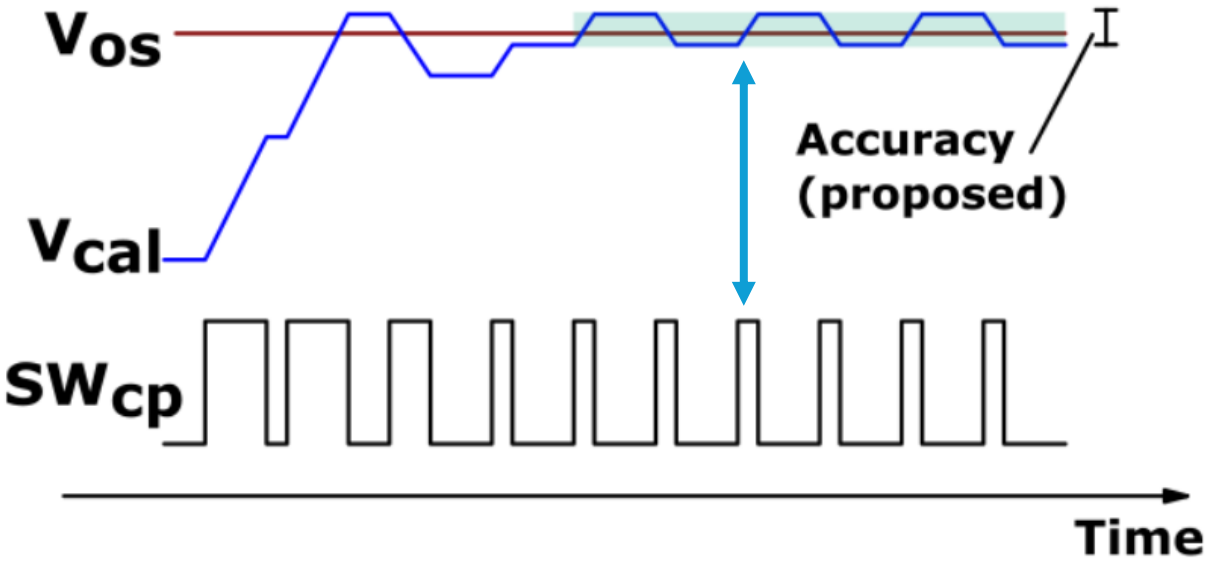

Vcm
Voffset
CAL
Vout+
Vin+
In+
CAL CAL
Vout−
Vin−
In−
CAL CAL
Mc2
Icp
Vcm
VC
Vb
Mc1
CH
CAL
Icp


VC
Voffset'
1
Vout
0
CLK

Accuracy bottleneck

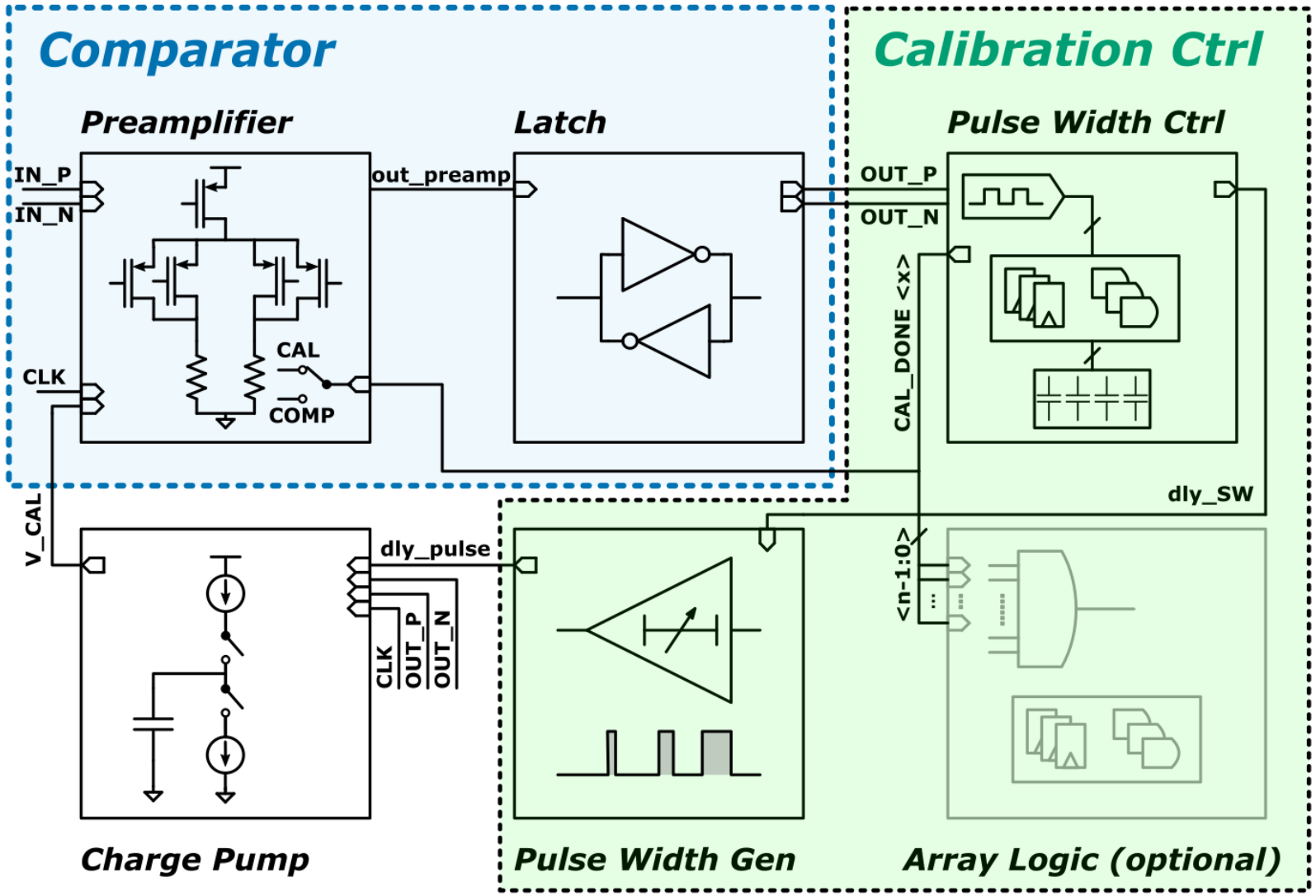


Accuracy limited
by CLK speed

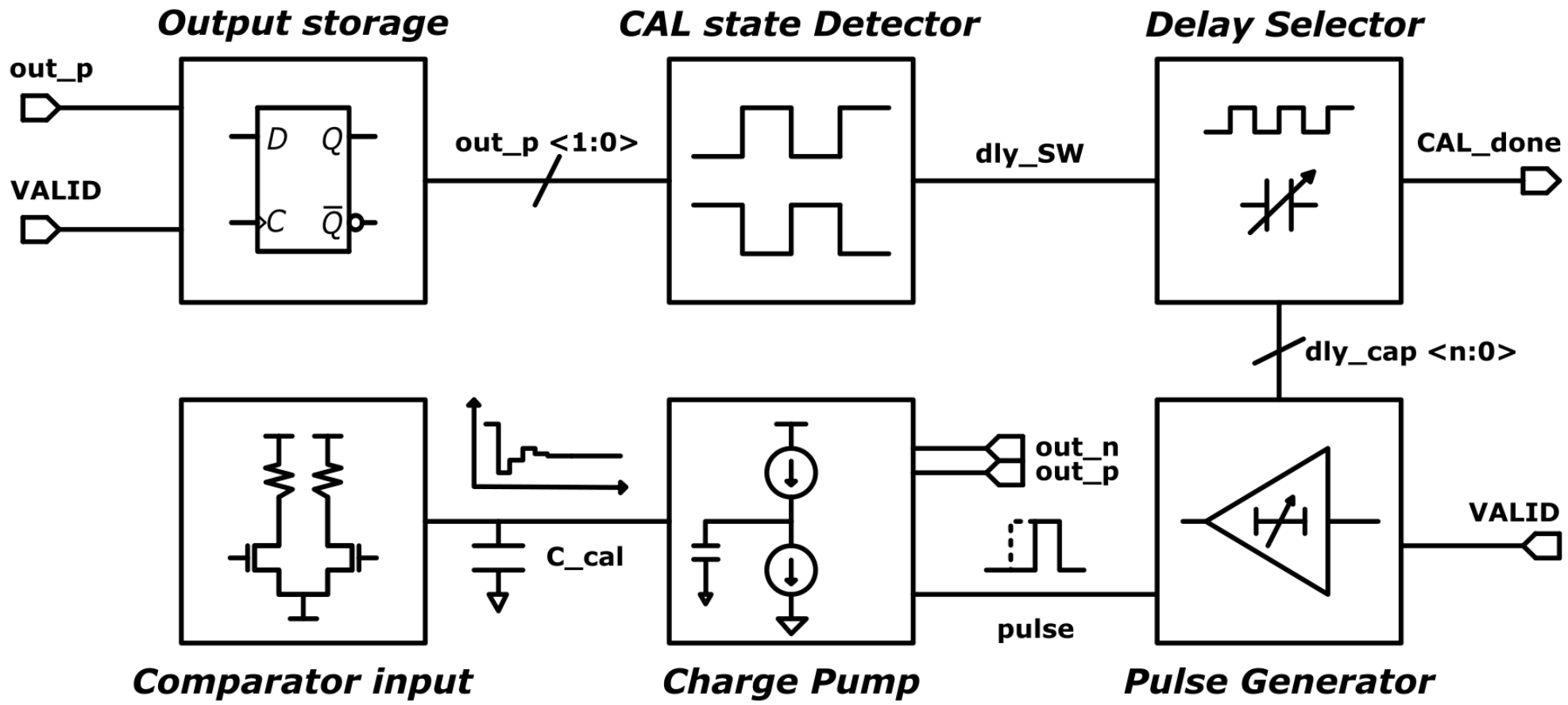


Pulse instead
of CLK

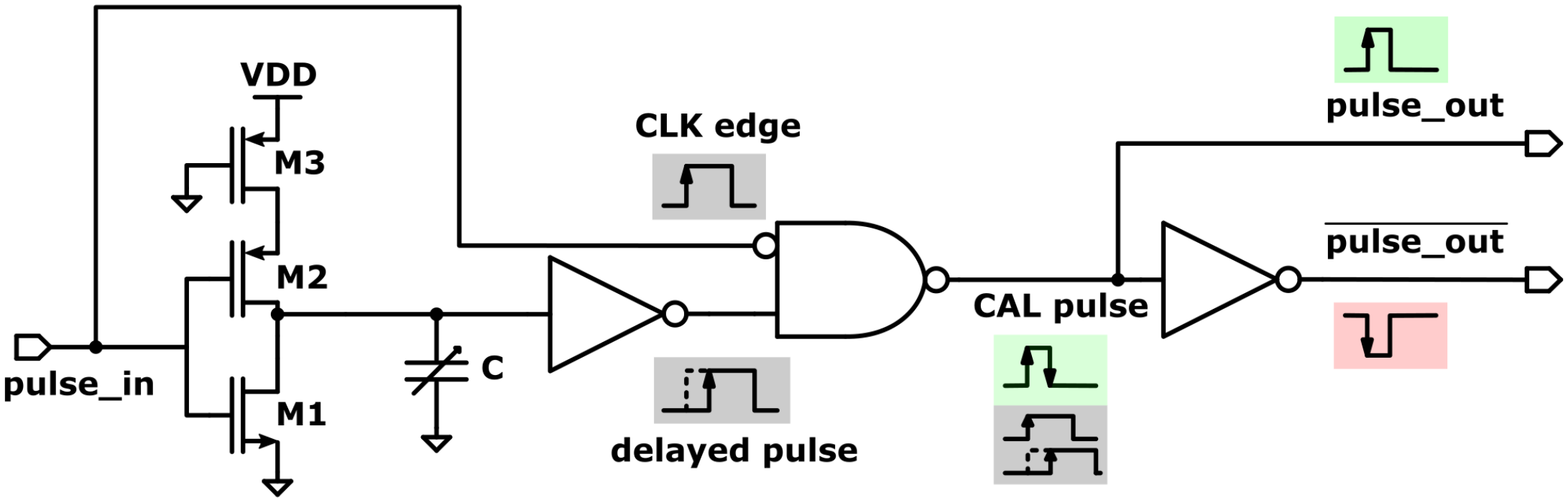
Proposed method



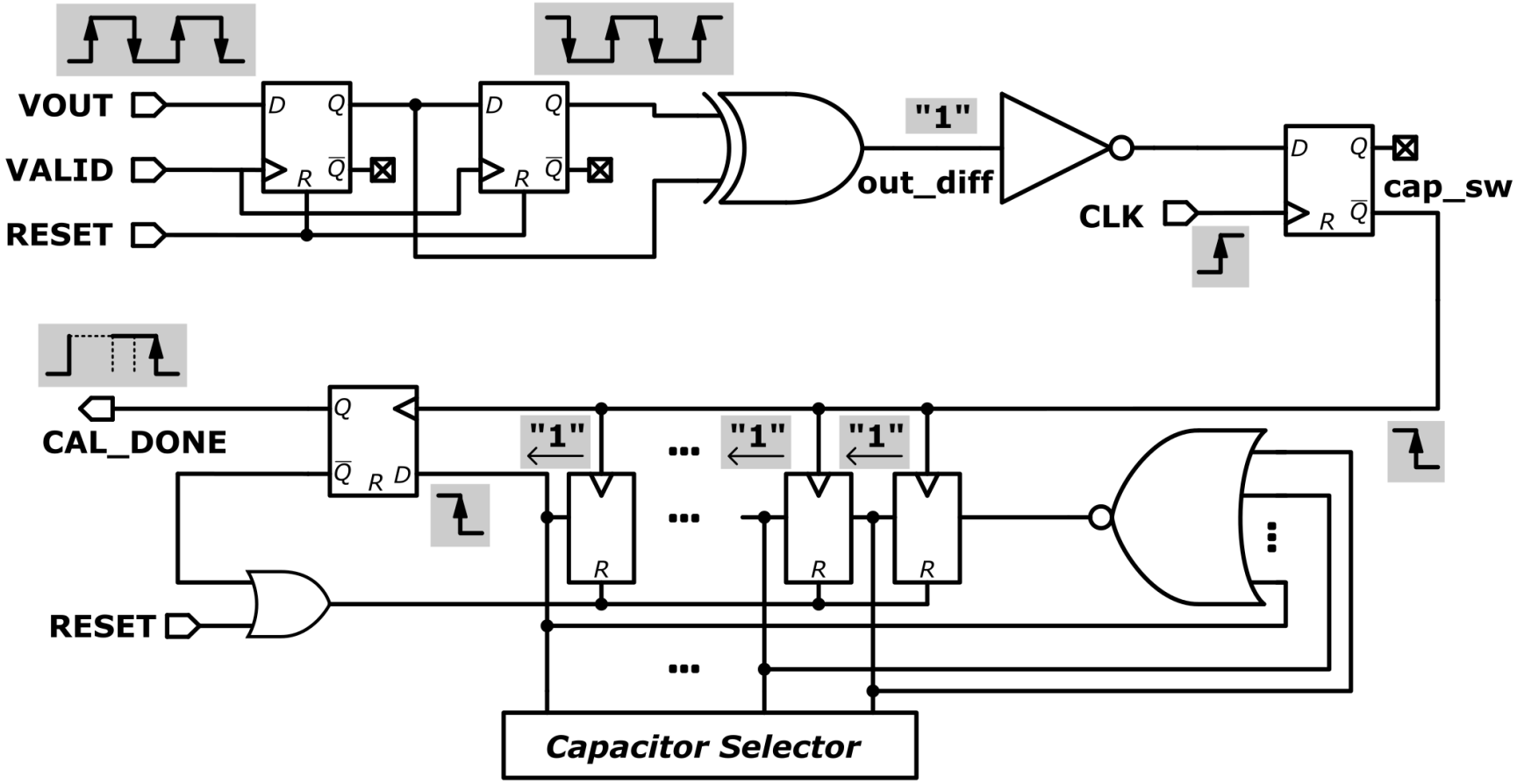
Calibration



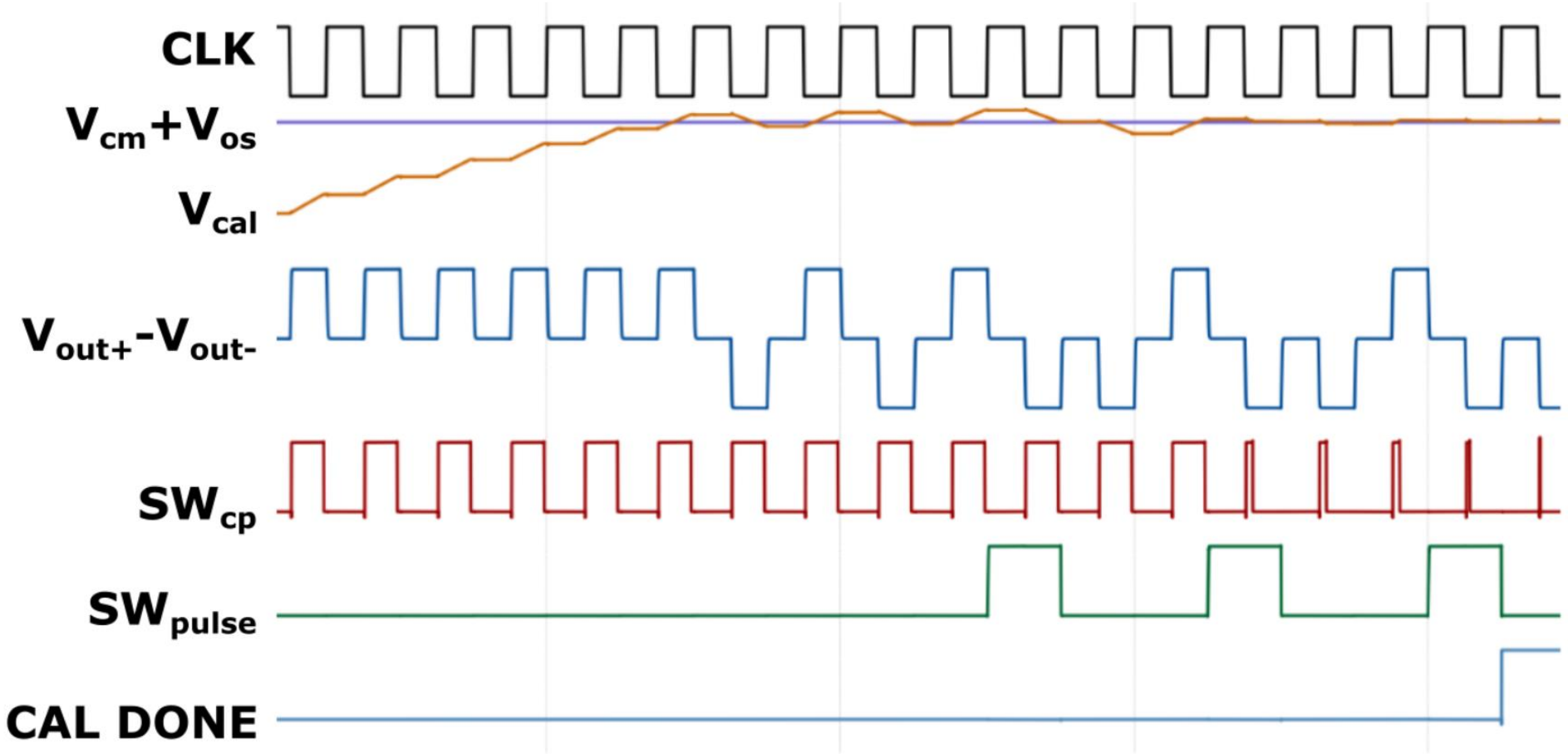
Pulse


VDD
M3
M2
M1
pulse_in
C
CLK edge
delayed pulse
CAL pulse
pulse_out
pulse_out

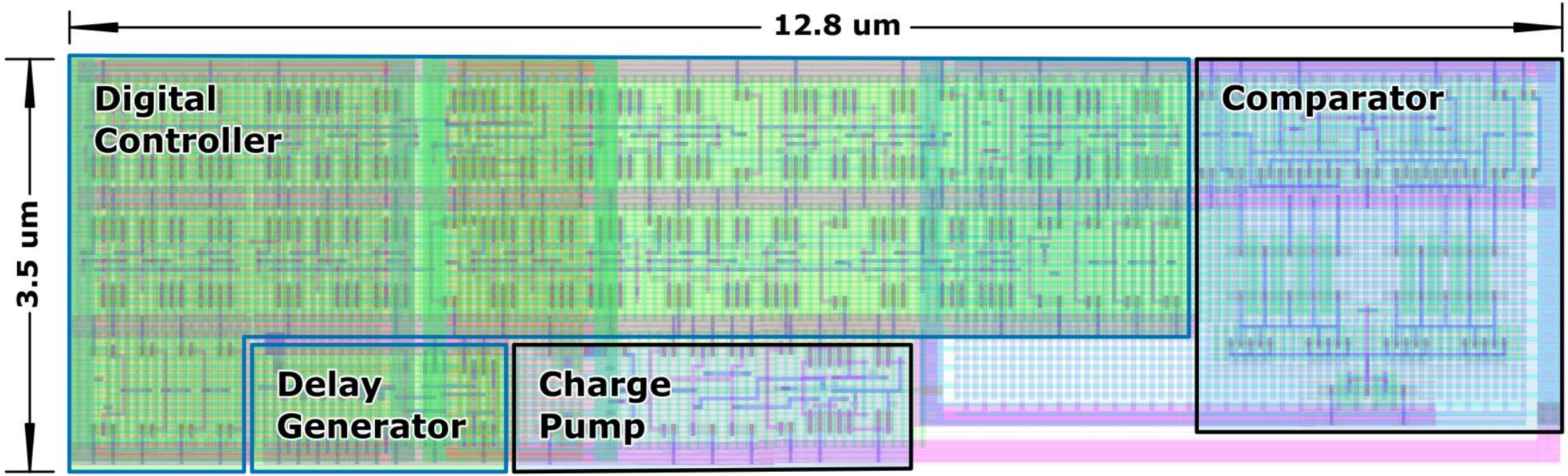
Control


VOUT
VALID
RESET
D
Q
R
$\overline{Q}$
"1"
out_diff
CLK
cap_sw
CAL_DONE
RESET
Capacitor Selector

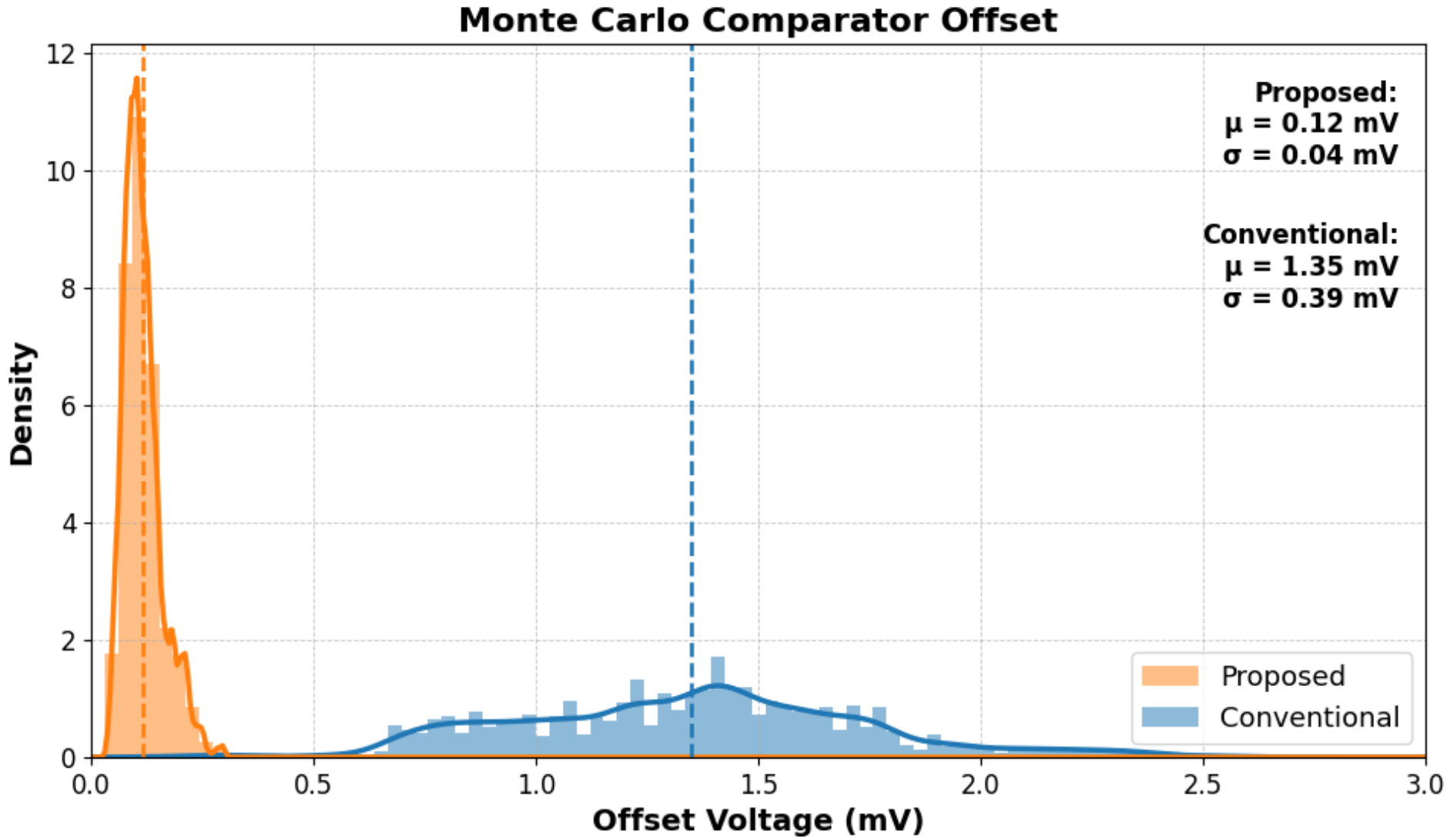
Timing



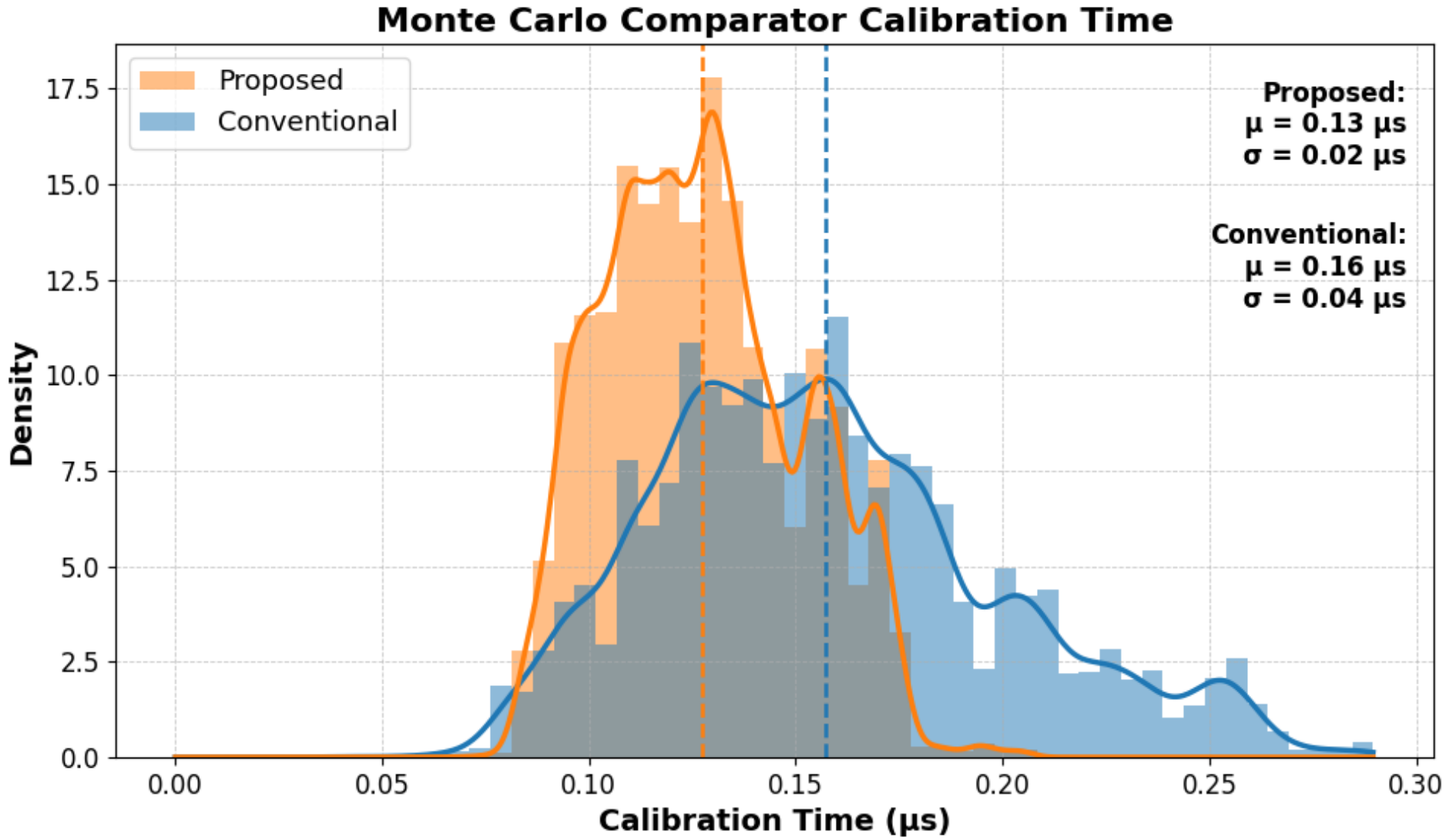
Layout


12.8 um
3.5 um
Digital Controller
Comparator
Delay Generator
Charge Pump

Result: offset



Result: calibration time



Result: comparison

	This work	ISCAS'25 [2]	ISCAS'25 [3]	ISCAS'20 [4]	ASSCC'11 [1]
Process	**12 nm**	12 nm	28 nm	14 nm	90 nm
Operation frequency	**50–500 MHz**	1.5 GHz	4 GHz	1 GHz	250 MHz
Offset	**120 µV**	1.9 mV	136 µV	259 µV	1.3 mV
Power	**10 µW**	240 µW	220 µW	213 µW	40 µW

References

[1] M. Miyahara, Y. Asada, D. Paik, and A. Matsuzawa, “A low-noise self-calibrating dynamic comparator for high-speed adcs,” in 2008 IEEE Asian Solid-State Circuits Conference, pp. 269–272, 2008.

[2] Yang et al.“A high-speed high-precision comparator with offset calibration for sha less pipeline adcs,” in 2025 IEEE International Symposium on Circuits and Systems (ISCAS)

[3] Xu et al. “A low-noise high-precision dual-mode offset calibration for high-speed dynamic comparators,” in 2025 IEEE International Symposium on Circuits and Systems (ISCAS)

[4] Lee et al. “An effective transconductance controlled offset calibration for dynamic comparators,” in Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS)